Arquitectura de Robots

Robot Scara (DH) Method

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1 Denavit-Hartenberg SCARA

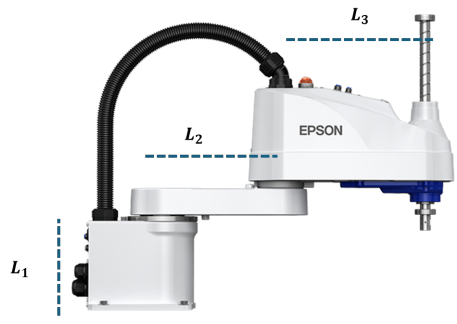




Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

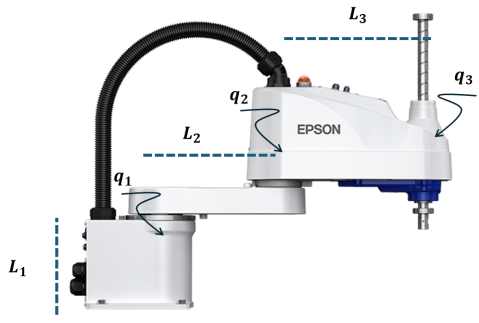
1. Number the links starting with 1 (first moving link in the chain) and ending with n (last moving link). The fixed base of the robot shall be numbered as link 0.



Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

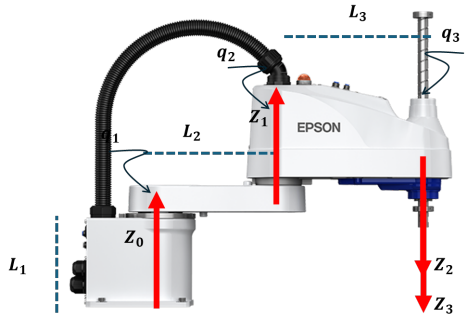
2. Number each joint starting with 1 (the one corresponding to the first degree of freedom) and ending in n.



Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

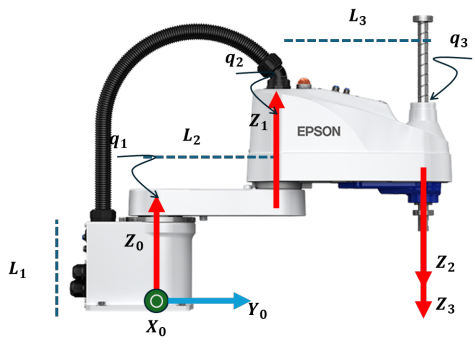
3. Locate the axis of each joint. If it is rotating, the axis will be its own axis of rotation; If it is prismatic, it will be the axis along which the displacement occurs.



Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

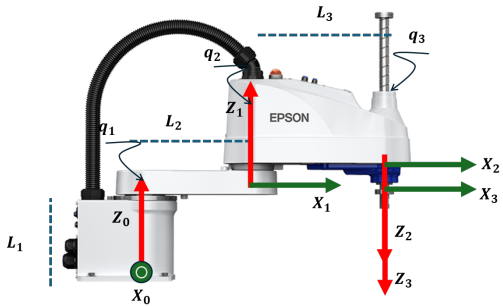
4. For i from 0 to $n-1$, place the axis z_i on the axis of the joint $i+1$.
5. Place the system origin of the base S_o at any point on the axis Z_o. The axes X_o and Y_o shall be so positioned as to form a right-handed system with Z_o



Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

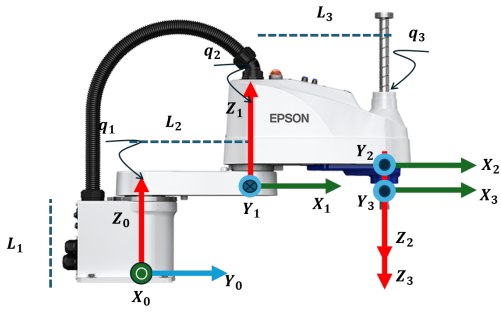
6. For i from 1 to $n-l$, locate the system S_i (solidarity to the link i) at the intersection of the axis Z_i with the normal line common to Z_{i-1} y Z_i
7. Place x_i on the common normal line to Z_{i+1} and Z_i.



Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

8. Place Y_i so that it forms a right-handed system with X_i and Z_i. (dextrorotatory).
9. Place the system S_n at the end of the robot so that Z_n matches the direction of Z_{n-1} and X_n is normal to Z_{n-1} and Z_n



Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Translation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1			
S_2				
S_3				

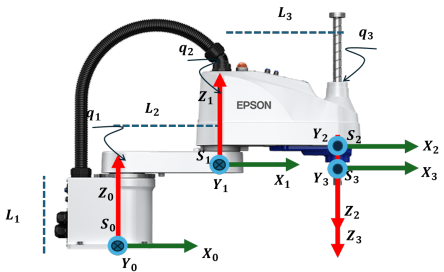


Figure 1: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1		
S_2				
S_3				

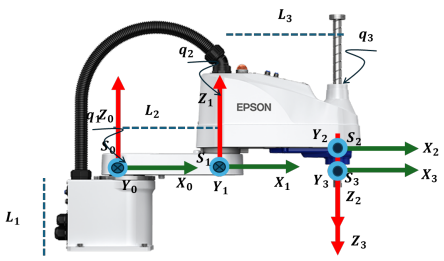


Figure 2: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	
S_2				
S_3				

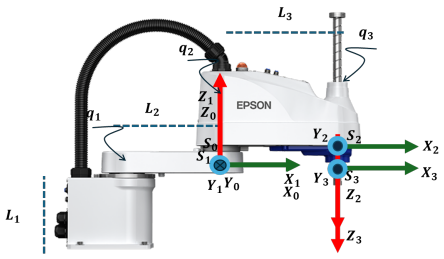


Figure 3: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2				
S_3				

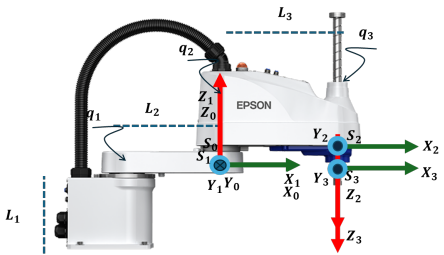


Figure 4: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2				
S_3				

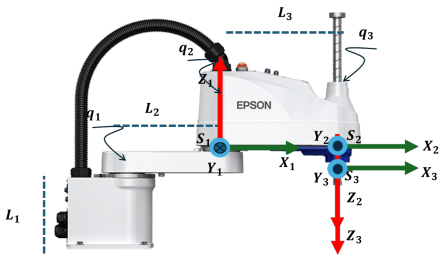


Figure 5: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2			
S_3				

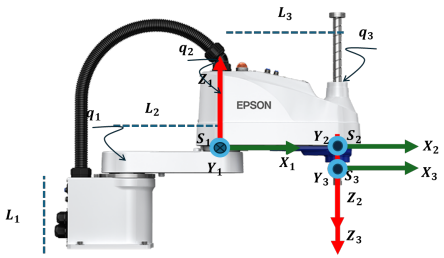


Figure 6: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0		
S_3				

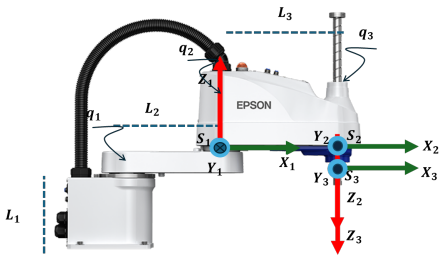


Figure 7: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	
S_3				

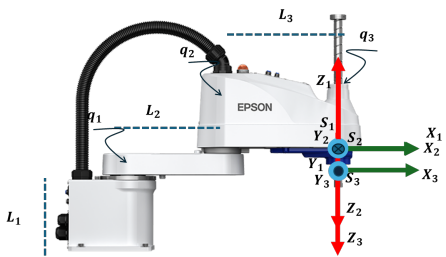


Figure 8: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3				

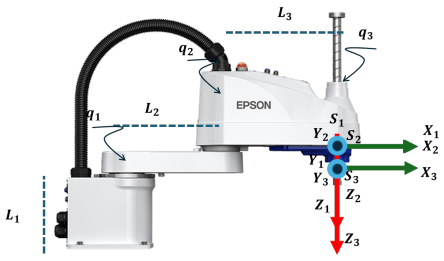


Figure 9: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3				

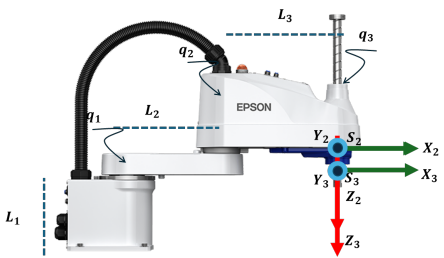


Figure 10: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3	0			

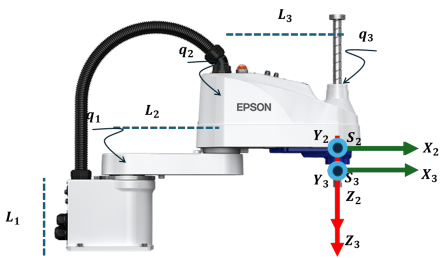


Figure 11: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3	0	q_3		

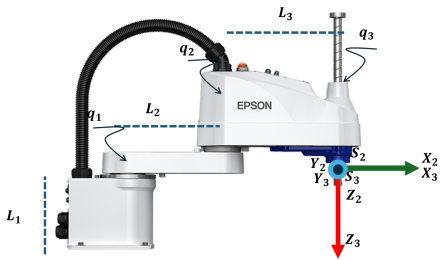


Figure 12: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3	0	q_3	0	

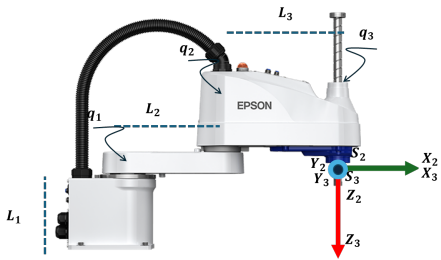


Figure 13: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3	0	q_3	0	0

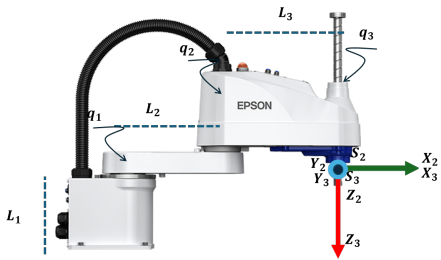


Figure 14: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

- θ_i: Rotation about z_{i-1}
- d_i: Translation along z_{i-1}
- a_i: Translation along x_i
- α_i: Rotation about x_i

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3	0	q_3	0	0

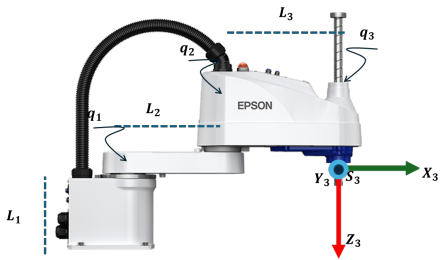


Figure 15: Robot 3DoF RRP

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3	0	q_3	0	0

$$ {}^{i-1}A_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ A_1 = \begin{bmatrix} cq_1 & -sq_1 & 0 & L_2 cq_1 \\ sq_1 & cq_1 & 0 & L_2 sq_1 \\ 0 & 0 & 1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1) $$

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3	0	q_3	0	0

$$ {}^{i-1}A_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ A_2 = \begin{bmatrix} cq_2 & sq_2 & 0 & L_3 cq_2 \\ sq_2 & -cq_2 & 0 & L_3 sq_2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2) $$

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

S_0	θ_i	d_i	a_i	α_i
S_1	q_1	L_1	L_2	$0°$
S_2	q_2	0	L_3	180
S_3	0	q_3	0	0

$$ {}^{i-1}A_i = \begin{bmatrix} \cos\theta_i & -\cos\alpha_i \sin\theta_i & \sin\alpha_i \sin\theta_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\alpha_i \cos\theta_i & -\sin\alpha_i \cos\theta_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ A_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3} $$

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

$$^{0}A_3 = ^{0}A_1 \cdot ^{1}A_2 \cdot ^{2}A_3$$

$$^{0}A_3 = \begin{bmatrix} cq_1 & -sq_1 & 0 & L_2cq_1 \\ sq_1 & cq_1 & 0 & L_2sq_1 \\ 0 & 0 & 1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} cq_2 & sq_2 & 0 & L_3cq_2 \\ sq_2 & -cq_2 & 0 & L_3sq_2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

$$
{}^{0}A_1 \cdot {}^{1}A_2 = \begin{bmatrix} cq_1 & -sq_1 & 0 & L_2cq_1 \\ sq_1 & cq_1 & 0 & L_2sq_1 \\ 0 & 0 & 1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} cq_2 & sq_2 & 0 & L_3cq_2 \\ sq_2 & -cq_2 & 0 & L_3sq_2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{0}A_1 \cdot {}^{1}A_2 = \begin{bmatrix} cq_1cq_2 - sq_1sq_2 & cq_1sq_2 + sq_1cq_2 & 0 & L_2cq_1 + L_3(cq_1cq_2 - sq_1sq_2) \\ sq_1cq_2 + cq_1sq_2 & sq_1sq_2 - cq_1cq_2 & 0 & L_2sq_1 + L_3(sq_1cq_2 + cq_1sq_2) \\ 0 & 0 & -1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5}
$$

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

$$\cos(a+b) = \cos a \cos b - \sin a \sin b \tag{6}$$

$$\sin(a+b) = \sin a \cos b + \cos a \sin b \tag{7}$$

$$cq_1 cq_2 - sq_1 sq_2 = c(q_1 + q_2)$$

$$cq_1 sq_2 + sq_1 cq_2 = s(q_1 + q_2)$$

$${}^0A_2 \cdot {}^2A_3 = \begin{bmatrix} c(q_1+q_2) & s(q_1+q_2) & 0 & L_2c(q_1) + L_3c(q_1+q_2) \\ s(q_1+q_2) & -c(q_1+q_2) & 0 & L_2s(q_1) + L_3s(q_1+q_2) \\ 0 & 0 & -1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{8}$$

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

$$
{}^{0}A_{2} \cdot {}^{2}A_{3} = \begin{bmatrix} c(q_1+q_2) & s(q_1+q_2) & 0 & L_2c(q_1)+L_3c(q_1+q_2) \\ s(q_1+q_2) & -c(q_1+q_2) & 0 & L_2s(q_1)+L_3s(q_1+q_2) \\ 0 & 0 & -1 & L_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^{0}A_{3} = \begin{bmatrix} c(q_1+q_2) & s(q_1+q_2) & 0 & L_2c(q_1)+L_3c(q_1+q_2) \\ s(q_1+q_2) & -c(q_1+q_2) & 0 & L_2s(q_1)+L_3s(q_1+q_2) \\ 0 & 0 & -1 & L_1-q_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{9}
$$

Scara LS6-B, DH Method

1 Denavit-Hartenberg SCARA

Position :

$$P_x = L_2 c(q_1) + L_3 c(q_1 + q_2)$$

$$P_y = L_2 s(q_1) + L_3 s(q_1 + q_2)$$

$$P_z = L_1 - q_3$$

Orientation:

$${}^3R_0 = [\mathbf{n}, \mathbf{o}, \mathbf{a}]$$

$${}^0R_3 = \begin{bmatrix} c(q_1+q_2) & s(q_1+q_2) & 0 \\ s(q_1+q_2) & -c(q_1+q_2) & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

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Q&A

Thank you for your attention

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